

Name: Prashanth Raghavendra Rao

Task: Terraform 1,2

Date: 16/07/2025

Task Description:

1. Launch Linux EC2 instances in two regions using a single Terraform file.
2. Create 2 EC2 instances on 2 different regions and install nginx using terraform script.

```
terraform-ec2-nginx -- zsh — 248x83

(base) prashanth@Prashanth-MacBook-Pro-2 Terraform % terraform --version
Terraform v1.12.2
on darwin_arm64
(base) prashanth@Prashanth-MacBook-Pro-2 Terraform % aws configure
AWS Access Key ID [*****]:
AWS Secret Access Key [*****]:
Default region name [ap-south-1]:
Default output format [json]:
(base) prashanth@Prashanth-MacBook-Pro-2 Terraform % cd terraform-ec2-nginx
(base) prashanth@Prashanth-MacBook-Pro-2 terraform-ec2-nginx % nano main.tf
(base) prashanth@Prashanth-MacBook-Pro-2 terraform-ec2-nginx % cat main.tf
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
  required_version = ">= 1.3.0"
}

# Provider for Mumbai
provider "aws" {
  region = "ap-south-1"
  alias = "mumbai"
}

# Provider for Oregon
provider "aws" {
  region = "us-west-2"
  alias = "oregon"
}

# EC2 in Mumbai
resource "aws_instance" "mumbai_instance" {
  provider = aws.mumbai
  ami      = var.ami_mumbai
  instance_type = var.instance_type
  key_name = var.key_name

  user_data = <<-EOF
  #!/bin/bash
  sudo apt update -y
  sudo apt install nginx -y
  sudo systemctl enable nginx
  sudo systemctl start nginx
  EOF

  tags = {
    Name = "Nginx-Mumbai"
  }
}

# EC2 in Oregon
resource "aws_instance" "oregon_instance" {
  provider = aws.oregon
  ami      = var.ami_oregon
  instance_type = var.instance_type
  key_name = var.key_name

  user_data = <<-EOF
  #!/bin/bash
  sudo apt update -y
  sudo apt install nginx -y
  sudo systemctl enable nginx
  sudo systemctl start nginx
  EOF

  tags = {
    Name = "Nginx-Oregon"
  }
}

(base) prashanth@Prashanth-MacBook-Pro-2 terraform-ec2-nginx %
```

terraform-ec2-nginx — zsh — 166x46

```
(base) prashanthr@Prashanth's-MacBook-Pro-2 terraform-ec2-nginx % nano variables.tf
(base) prashanthr@Prashanth's-MacBook-Pro-2 terraform-ec2-nginx % cat variables.tf
variable "ami_mumbai" {
  description = "Ubuntu AMI in ap-south-1"
  default     = "ami-0f918f7e67a3323f0"
}

variable "ami_oregon" {
  description = "Ubuntu AMI in us-west-2"
  default     = "ami-05f991c49d264708f"
}

variable "instance_type" {
  description = "Instance type"
  default     = "t2.micro"
}

variable "key_name" {
  description = "PrashanthEC2"
  default     = "PrashanthEC2.pem"
}

(base) prashanthr@Prashanth's-MacBook-Pro-2 terraform-ec2-nginx % nano outputs.tf
(base) prashanthr@Prashanth's-MacBook-Pro-2 terraform-ec2-nginx % cat outputs.tf
output "mumbai_public_ip" {
  value = aws_instance.mumbai_instance.public_ip
}

output "oregon_public_ip" {
  value = aws_instance.oregon_instance.public_ip
}

(base) prashanthr@Prashanth's-MacBook-Pro-2 terraform-ec2-nginx %
```

terraform-ec2-nginx — zsh — 166x46

```
(base) prashanthr@Prashanth's-MacBook-Pro-2 terraform-ec2-nginx % nano variables.tf
(base) prashanthr@Prashanth's-MacBook-Pro-2 terraform-ec2-nginx % cat variables.tf
variable "ami_mumbai" {
  description = "Ubuntu AMI in ap-south-1"
  default     = "ami-0f918f7e67a3323f0"
}

variable "ami_oregon" {
  description = "Ubuntu AMI in us-west-2"
  default     = "ami-05f991c49d264708f"
}

variable "instance_type" {
  description = "Instance type"
  default     = "t2.micro"
}

variable "key_name" {
  description = "PrashanthEC2"
  default     = "PrashanthEC2"
}

(base) prashanthr@Prashanth's-MacBook-Pro-2 terraform-ec2-nginx % cat outputs.tf
output "mumbai_public_ip" {
  value = aws_instance.mumbai_instance.public_ip
}

output "oregon_public_ip" {
  value = aws_instance.oregon_instance.public_ip
}

(base) prashanthr@Prashanth's-MacBook-Pro-2 terraform-ec2-nginx %
```

```

(base) prashanthr@Prashanth-MacBook-Pro-2 terraform-ec2-nginx % terraform init

Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v5.100.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
(base) prashanthr@Prashanth-MacBook-Pro-2 terraform-ec2-nginx % terraform plan
aws_instance.mumbai_instance: Refreshing state... [id=i-02af5b3315097f3ae]

Note: Objects have changed outside of Terraform

Terraform detected the following changes made outside of Terraform since the last "terraform apply" which may have affected this plan:

# aws_instance.mumbai_instance has been deleted
- resource "aws_instance" "mumbai_instance" {
  id           = "i-02af5b3315097f3ae"
  public_ip    = "52.66.88.189" -> null
  tags         = {
    "Name" = "Nginx-Mumbai"
  }
  # (39 unchanged attributes hidden)
  # (8 unchanged blocks hidden)
}

Unless you have made equivalent changes to your configuration, or ignored the relevant attributes using ignore_changes, the following plan may include actions to undo or respond to these changes.

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.mumbai_instance will be created
+ resource "aws_instance" "mumbai_instance" {
  + ami              = "ami-0f918f7e67a3323f0"
  + arn              = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone = (known after apply)
  + cpu_core_count    = (known after apply)
  + cpu_threads_per_core = (known after apply)
  + disable_api_stop   = (known after apply)
  + ebs_block_device  = (known after apply)
  + enclave_options   = (known after apply)
  + ephemeral_block_device = (known after apply)
  + instance_market_options = (known after apply)
  + maintenance_options = (known after apply)
  + metadata_options  = (known after apply)
  + network_interface = (known after apply)
  + private_dns_name_options = (known after apply)
  + root_block_device = (known after apply)
}

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
~ mumbai_public_ip = "52.66.88.189" -> (known after apply)
+ oregon_public_ip = (known after apply)

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
(base) prashanthr@Prashanth-MacBook-Pro-2 terraform-ec2-nginx %
(base) prashanthr@Prashanth-MacBook-Pro-2 terraform-ec2-nginx %
(base) prashanthr@Prashanth-MacBook-Pro-2 terraform-ec2-nginx % terraform apply
aws_instance.mumbai_instance: Refreshing state... [id=i-02af5b3315097f3ae]

Note: Objects have changed outside of Terraform

Terraform detected the following changes made outside of Terraform since the last "terraform apply" which may have affected this plan:

# aws_instance.mumbai_instance has been deleted
- resource "aws_instance" "mumbai_instance" {
  id           = "i-02af5b3315097f3ae"
  public_ip    = "52.66.88.189" -> null
  tags         = {
    "Name" = "Nginx-Mumbai"
  }
  # (39 unchanged attributes hidden)
  # (8 unchanged blocks hidden)
}

Unless you have made equivalent changes to your configuration, or ignored the relevant attributes using ignore_changes, the following plan may include actions to undo or respond to these changes.
```

```
terraform-ec2-nginx -- -zsh -- 172x55

+ user_data_replace_on_change = false
+ vpc_security_group_ids = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)
}

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
- mumbai_public_ip = "52.66.88.189" -> (known after apply)
+ oregon_public_ip = (known after apply)

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_instance.mumbai_instance: Creating...
aws_instance.oregon_instance: Creating...
aws_instance.mumbai_instance: Still creating... [00m10s elapsed]
aws_instance.oregon_instance: Still creating... [00m10s elapsed]
aws_instance.mumbai_instance: Still creating... [00m20s elapsed]
aws_instance.oregon_instance: Still creating... [00m20s elapsed]
aws_instance.oregon_instance: Creation complete after 27s [id=i-0e5da204623627f55]
aws_instance.mumbai_instance: Still creating... [00m30s elapsed]
aws_instance.mumbai_instance: Creation complete after 32s [id=i-01795c4a783830737]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:
mumbai_public_ip = "52.66.141.118"
oregon_public_ip = "54.282.7.240"
```

EC2 Global View

Region explorer

Global search

Updated 1 minute ago

Global search (1/3)

Perform a global search to search for specific resources across all Regions for which your account is enabled

Download CSV

Manage

Find resources by attribute or tag

Resource Type = Instance

Clear Filters

	Name	Resource ID	Resource Type	Region
☐	Ngixn-Mumbai	i-02af5b3315097f3ae	Instance	ap-sout...
☒	Ngixn-Mumbai	i-01795c4a783830737	Instance	ap-sout...
☐	Ngixn-Oregon	i-0e5da204623627f55	Instance	us-west-2

i-01795c4a783830737 (Ngixn-Mumbai) - ap-south-1

Details

Security groups

Volumes

Tags

Instance ID

[i-01795c4a783830737 \(Ngixn-Mumbai\)](#)

Public IPv4 address

[52.66.141.118](#)

Private IPv4 addresses

[172.31.9.221](#)

Public IPv4 DNS

[ec2-52-66-141-118.ap-south-1.compute.amazonaws.com](#) [open address](#)

Instance type

t2.micro

VPC ID

[vpc-032038c199af20b71](#)

Instance state

[Running](#)

Private IPv4 DNS

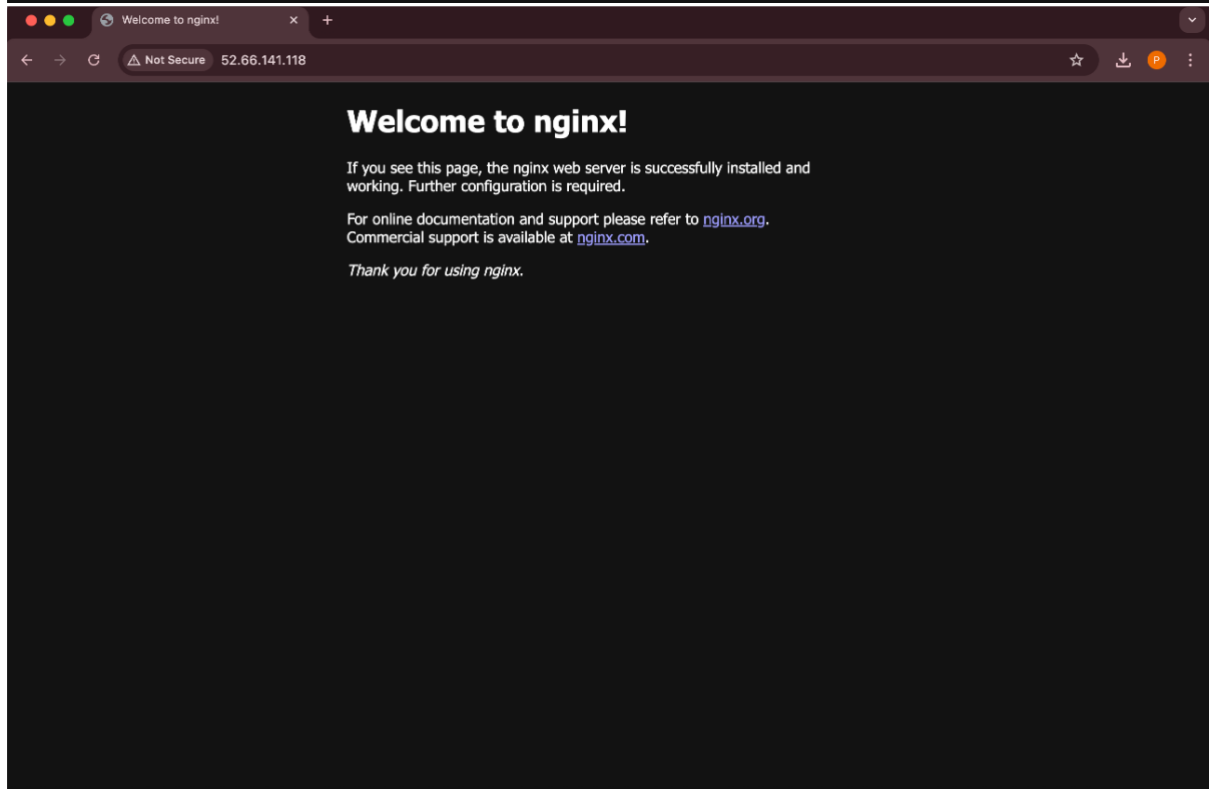
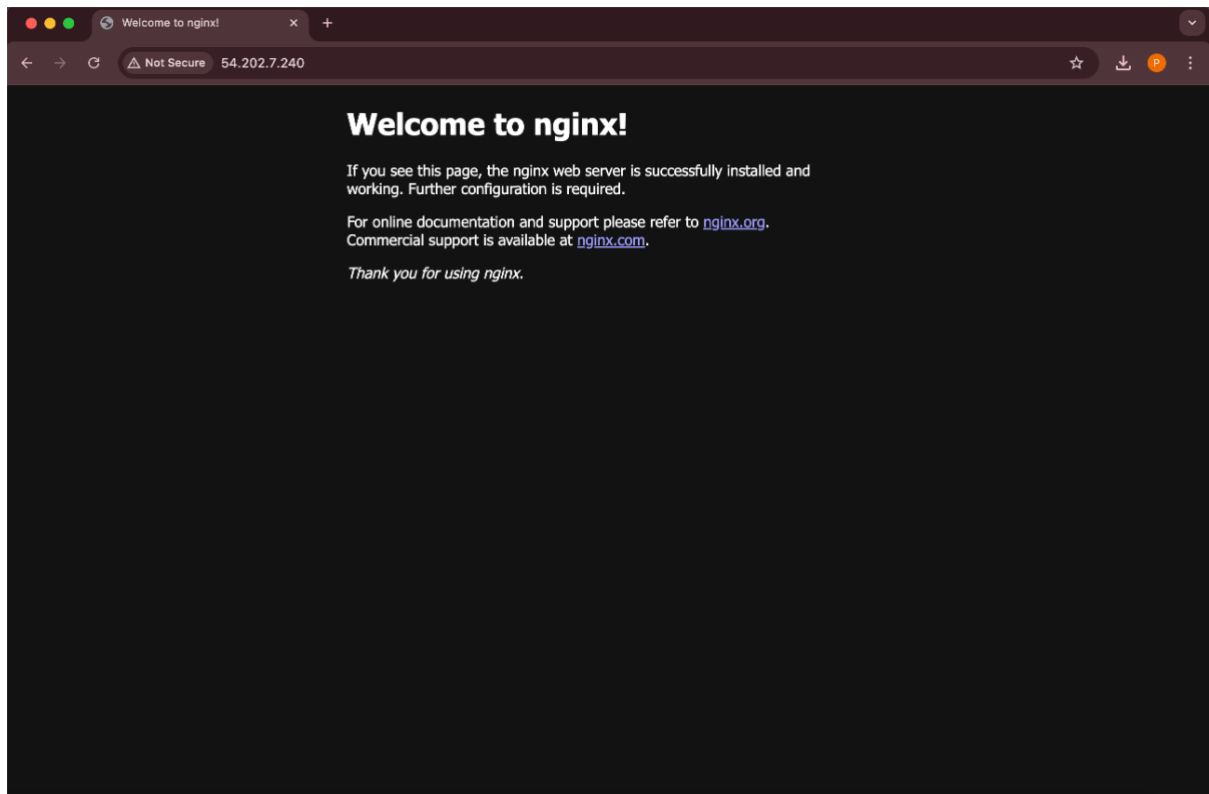
[ip-172-31-9-221.ap-south-1.compute.internal](#)

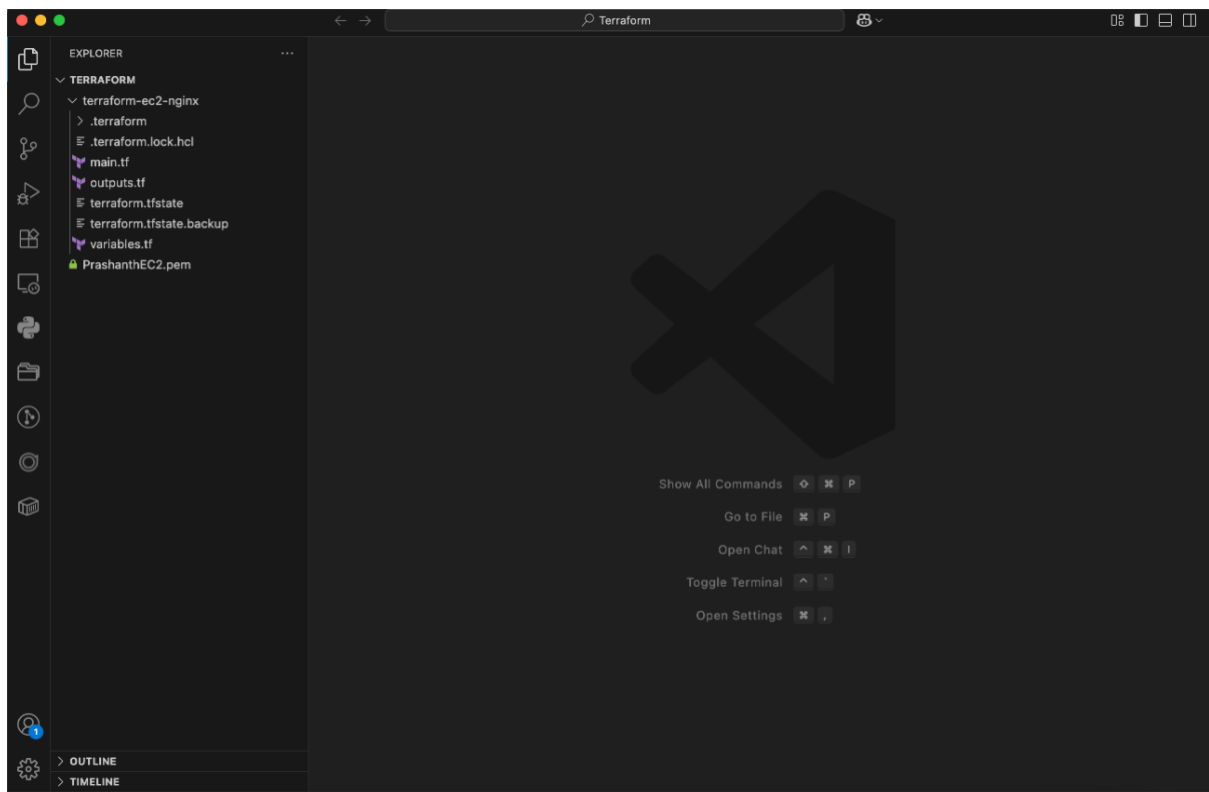
Launch Time

2025/07/16 01:22 GMT+5:30 (3 minutes)

Subnet ID

[subnet-0f24967078e8af115](#)





```
- metadata_options {
  - http_endpoint           = "enabled" -> null
  - http_protocol_ipv6      = "disabled" -> null
  - http_put_response_hop_limit = 2 -> null
  - http_tokens             = "required" -> null
  - instance_metadata_tags   = "disabled" -> null
}

- private_dns_name_options {
  - enable_resource_name_dns_a_record    = false -> null
  - enable_resource_name_dns_aaaa_record = false -> null
  - hostname_type                        = "ip-name" -> null
}

- root_block_device {
  - delete_on_termination = true -> null
  - device_name           = "/dev/sdal" -> null
  - encrypted             = false -> null
  - iops                  = 3000 -> null
  - tags                  = {} -> null
  - tags_all              = {} -> null
  - throughput            = 125 -> null
  - volume_id             = "vol-027361884678b666b" -> null
  - volume_size           = 8 -> null
  - volume_type           = "gp3" -> null
  # (1 unchanged attribute hidden)
}
}

Plan: 0 to add, 0 to change, 2 to destroy.

Changes to Outputs:
- mumbai_public_ip = "52.66.141.118" -> null
- oregon_public_ip = "54.202.7.240" -> null

Do you really want to destroy all resources?
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

aws_instance.mumbai_instance: Destroying... [id=i-01795c4a783830737]
aws_instance.oregon_instance: Destroying... [id=i-0e5da204623627f55]
aws_instance.mumbai_instance: Still destroying... [id=i-01795c4a783830737, 00m10s elapsed]
aws_instance.oregon_instance: Still destroying... [id=i-0e5da204623627f55, 00m10s elapsed]
aws_instance.mumbai_instance: Still destroying... [id=i-01795c4a783830737, 00m20s elapsed]
aws_instance.oregon_instance: Still destroying... [id=i-0e5da204623627f55, 00m20s elapsed]
aws_instance.mumbai_instance: Still destroying... [id=i-01795c4a783830737, 00m30s elapsed]
aws_instance.oregon_instance: Still destroying... [id=i-0e5da204623627f55, 00m30s elapsed]
aws_instance.mumbai_instance: Destruction complete after 30s
aws_instance.oregon_instance: Destruction complete after 33s

Destroy complete! Resources: 2 destroyed.
(base) prashanth@Prashanth-MacBook-Pro-2 terraform-ec2-nginx %
```

```
terraform-ec2-nginx -- zsh -- 172x65
aws_instance.oregon_instance: Still creating... [00m10s elapsed]
aws_instance.mumbai_instance: Still creating... [00m20s elapsed]
aws_instance.oregon_instance: Still creating... [00m20s elapsed]
aws_instance.oregon_instance: Creation complete after 27s [id=i-0e5da204623627f55]
aws_instance.mumbai_instance: Still creating... [00m30s elapsed]
aws_instance.mumbai_instance: Creation complete after 32s [id=i-01795c4a783830737]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:
mumbai_public_ip = "52.66.141.118"
oregon_public_ip = "54.202.7.240"
(base) prashanth@Prashanth-MacBook-Pro-2: terraform-ec2-nginx %
(base) prashanth@Prashanth-MacBook-Pro-2: terraform-ec2-nginx %
(base) prashanth@Prashanth-MacBook-Pro-2: terraform-ec2-nginx % terraform destroy
aws_instance.mumbai_instance: Refreshing state... [id=i-01795c4a783830737]
aws_instance.oregon_instance: Refreshing state... [id=i-0e5da204623627f55]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_instance.mumbai_instance will be destroyed
- resource "aws_instance" "mumbai_instance" {
  ami           = "ami-0f918f7e67a3323f0" -> null
  arn           = "arn:aws:ec2:ap-south-1:235391889675:instance/i-01795c4a783830737" -> null
  associate_public_ip_address = true -> null
  availability_zone = "ap-south-1b" -> null
  cpu_core_count   = 1 -> null
  cpu_threads_per_core = 1 -> null
  disable_api_stop  = false -> null
  disable_api_termination = false -> null
  ebs_optimized     = false -> null
  get_password_data = false -> null
  hibernation       = false -> null
  id               = "i-01795c4a783830737" -> null
  instance_initiated_shutdown_behavior = "stop" -> null
  instance_state    = "running" -> null
  instance_type     = "t2.micro" -> null
  ipv6_address_count = 0 -> null
  ipv6_addresses    = [] -> null
  key_name          = "PrashanthEC2" -> null
  monitoring        = false -> null
  placement_partition_number = 0 -> null
  primary_network_interface_id = "eni-033d24c72794a27f2" -> null
  private_dns       = "ip-172-31-9-221.ap-south-1.compute.internal" -> null
  private_ip        = "172.31.9.221" -> null
  public_dns        = "ec2-52-66-141-118.ap-south-1.compute.amazonaws.com" -> null
  public_ip         = "52.66.141.118" -> null
  secondary_private_ips = [] -> null
  security_groups   = [
    "sg-0a5a1a1a"
  ] -> null
}
```

EC2 Global View

Region explorer

Global search

Updated 1 minute ago

Global search (1/3)

Perform a global search to search for specific resources across all Regions for which your account is enabled

Download CSV

Manage

Find resources by attribute or tag

Resource Type = Instance

Clear filters

< 1 >

Name	Resource ID	Resource Type	Region
☐ Nginx-Mumbai	i-01795c4a783830737	Instance	ap-sout..
☒ Nginx-Mumbai	i-02af5b3315097f3ae	Instance	ap-sout..
☐ Nginx-Oregon	i-0e5da204623627f55	Instance	us-west-2

i-02af5b3315097f3ae (Nginx-Mumbai) - ap-south-1

Details

Security groups

Volumes

Tags

Instance ID

i-02af5b3315097f3ae (Nginx-Mumbai)

Public IPv4 address

-

Private IPv4 addresses

-

Instance state

Terminated

Public IPv4 DNS

-

Private IPv4 DNS

-

Instance type

t2.micro

Launch Time

2025/07/16 01:19 GMT+5:30 (16 minutes)

VPC ID

-

Subnet ID

-